

MIDTERM 1 REVIEW SHEET
MATH 411: HONORS COMPLEX VARIABLES, FALL 2026

Date: Friday, October 9, 2026, in class (CAB 369, 3:00–3:50 PM)
Length: 50 minutes
Format: Closed book. No notes, no textbook, no calculators or devices.
Coverage: Lectures 1–12; Lang §§1.1–1.7, 2.1–2.5, 3.2–3.3 (through Goursat)

Since the exam is closed book, you should be able to *state* the main definitions and theorems below precisely, not just recognize them. When you use a result from class in a solution, state clearly which result you are using and check that its hypotheses hold.

Lecture 13 (local primitives, Cauchy’s theorem on a disc) and Lecture 14 (computing real integrals) will *not* be tested on this midterm.

TOPICS

1. Basic complex numbers (Lectures 1–3; Lang §§1.1–1.4).

- Arithmetic in $\mathbb{C}$: real and imaginary parts, the conjugate $\bar{z}$, the modulus $|z| = \sqrt{z\bar{z}}$, and the identities $|zw| = |z||w|$, $\overline{zw} = \bar{z}\bar{w}$, $z^{-1} = \bar{z}/|z|^2$.
- The triangle inequality $|z + w| \leq |z| + |w|$ and its reverse form $||z| - |w|| \leq |z - w|$.
- Polar form $z = re^{i\theta}$; multiplication multiplies moduli and adds arguments. The definition $e^{a+ib} = e^a(\cos b + i\sin b)$.
- Solving $z^n = w$: the n distinct n -th roots of a nonzero w , and roots of unity.
- Stereographic projection and the extended plane $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$.
- Topology of $\mathbb{C}$: open and closed discs, open and closed sets, interior, boundary and adherent points; limits and continuity of functions $f : S \rightarrow \mathbb{C}$ in ε – δ form.

2. $\mathbb{C}$ -differentiability (Lectures 3–5; Lang §§1.5–1.7).

- Definition of $f'(z_0)$ as a limit over $h \in \mathbb{C}$, and of *holomorphic* on an open set. The equivalent “linear approximation” form $f(z_0 + h) - f(z_0) = f'(z_0)h + h\sigma(h)$ with $\sigma(h) \rightarrow 0$.
- Differentiable implies continuous; sum, product, quotient and chain rules. Polynomials are entire; rational functions are holomorphic away from the zeros of the denominator.
- Total (real) differentiability of $F = (u, v) : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^2$; the Jacobian; continuous partials $\Rightarrow$ totally differentiable.
- **Cauchy–Riemann criterion:** $f = u + iv$ is $\mathbb{C}$ -differentiable at $z = x + iy$ if and only if $F = (u, v)$ is totally differentiable at (x, y) and $u_x = v_y$, $u_y = -v_x$ there. In that case $f' = u_x + iv_x$. Know the proof, not just the statement.
- Using CR to show a function is (or is not) holomorphic; e.g. e^z is entire, $\bar{z}$ and $|z|^2$ are not holomorphic.
- On a connected open set, $f' \equiv 0$ implies f is constant (via “locally constant”). Typical consequences: if f is holomorphic on a disc and $\operatorname{Re} f$, $\operatorname{Im} f$ or $|f|$ is constant, then f is constant.
- Conformality: if $f'(z_0) \neq 0$, then f preserves oriented angles between curves at z_0 .

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- Green's theorem and the resulting first version of Cauchy's theorem (assuming f has continuous partials).

3. Power series (Lectures 6–9; Lang §§2.1–2.2).

- Formal power series $\mathbb{C}[[T]]$: addition, multiplication (Cauchy product), inversion (f is invertible iff $a_0 \neq 0$), composition $f \circ g$ when g has no constant term. Be able to compute the first few coefficients of $1/f$, $f \cdot g$, $f \circ g$.
- Convergence of sequences and series in $\mathbb{C}$; Cauchy sequences; absolute convergence; the comparison test.
- Sequences and series of functions: pointwise vs. uniform convergence, the sup norm $\|\cdot\|_S$, uniformly Cauchy sequences. A uniform limit of continuous functions is continuous. The comparison test for series of functions (Weierstrass M -test).
- **Radius of convergence:** every $\sum a_n z^n$ has $0 \leq r \leq \infty$ such that it converges absolutely on $D(0, r)$, uniformly on each closed disc $\overline{D}(0, s)$ with $s < r$, and diverges for $|z| > r$.
- **limsup and Hadamard's formula** $1/r = \limsup_n |a_n|^{1/n}$. Be able to compute radii of convergence (including series with gaps, such as $\sum a_n z^{n^2}$), and know the proof.
- Behaviour on the boundary circle can vary: compare $\sum n z^n$, $\sum z^n/n$, $\sum z^n/n^2$.
- The formal derivative $\sum n a_n z^{n-1}$ has the same radius of convergence, and a power series can be differentiated term by term inside its disc of convergence; hence it is infinitely $\mathbb{C}$ -differentiable there.

4. Analytic functions (Lectures 9–11; Lang §§2.3–2.5).

- Definition: f is *analytic* on U if near each $z_0 \in U$ it is given by a convergent power series in $z - z_0$.
- A power series with radius r is analytic on all of $D(0, r)$ (re-centring a power series).
- Analytic $\Rightarrow$ holomorphic, in fact infinitely $\mathbb{C}$ -differentiable; the Taylor coefficients are $a_k = f^{(k)}(z_0)/k!$. Sums and products of analytic functions are analytic.
- Uniqueness of power series expansions; **isolated zeros**; the **identity principle** (two convergent power series that agree at a sequence of nonzero points tending to 0 are the same power series).
- Analytic extensions of real functions, and uniqueness of such extensions; the complex exponential, sin, cos.

5. Path integrals and the Cauchy–Goursat theorem (Lectures 11–12; Lang §§3.2–3.3).

- Integrals of complex-valued functions of a real variable, and $|\int_a^b F| \leq \int_a^b |F|$.
- Piecewise C^1 paths and the path integral $\int_\gamma f dz = \int_a^b f(\gamma(t)) \gamma'(t) dt$. Be able to compute integrals over line segments, circles and rectangles (e.g. $\int_{|z|=1} z^n dz$ for all $n \in \mathbb{Z}$).
- Independence of parametrization; reversing orientation changes the sign.
- **The ML-estimate:** $|\int_\gamma f| \leq (\sup_\gamma |f|) \cdot L(\gamma)$.
- For a polynomial Q and a closed path γ , $\int_\gamma Q = 0$.
- **Cauchy–Goursat:** if f is holomorphic on an open set U and $R \subseteq U$ is a closed rectangle, then $\int_{\partial R} f = 0$. Know the structure of the proof: subdivide into four rectangles, pick the worst one, get nested rectangles shrinking to a point z_0 , use the linear approximation of f at z_0 together with the lemma on polynomials, and finish with the ML-estimate. Note that no continuity of f' is assumed (compare with the Green's theorem version).

HOW TO PREPARE

- Work through Problem Sets 1 and 2.

- Try the 2021 and 2022 Midterm 1 papers (posted with solutions on the course site) under exam conditions: 50 minutes, nothing on your desk. Note that the 2022 exam allowed a page of notes; this one does not.
- Practise writing out the key statements from memory: the definition of $\mathbb{C}$ -differentiability, the Cauchy–Riemann criterion, Hadamard’s formula, the definition of analytic, the identity principle, the ML-estimate, and Cauchy–Goursat.

PRACTICE PROBLEMS

These are for practice only; solutions will not be collected.

- (1) Find all solutions of $z^4 = -16$ and plot them. Then describe geometrically the set $\{z \in \mathbb{C} : |z - 1| = |z + i|\}$.
- (2) Show that if $|w| < 1$ and $|z| < 1$, then $\left| \frac{z - w}{1 - \bar{w}z} \right| < 1$. (*Hint: compare $|1 - \bar{w}z|^2 - |z - w|^2$.*)
- (3) Using the Cauchy–Riemann criterion, determine the set of points at which each function is $\mathbb{C}$ -differentiable:
 - (a) $f(z) = |z|^2$;
 - (b) $f(x + iy) = x^2 + iy^2$;
 - (c) $f(z) = z\bar{z}^2$.
- (4) Let f be holomorphic on the unit disc and suppose that $|f|$ is constant. Prove that f is constant.
- (5) Find the radius of convergence of
 - (a) $\sum_{n \geq 0} \frac{n^3}{3^n} z^n$;
 - (b) $\sum_{n \geq 0} 2^n z^{2n}$;
 - (c) $\sum_{n \geq 1} n! z^{n^2}$.
- (6) Find the terms of order ≤ 3 in the power series of $\frac{e^z}{1 - z}$ at $z = 0$, and of $\frac{1}{\cos z}$ at $z = 0$. What is the radius of convergence of the first?
- (7) Show directly from the definition that $f(z) = 1/z$ is analytic on $\mathbb{C} \setminus \{0\}$: expand it as a power series around an arbitrary $z_0 \neq 0$ and find the radius of convergence.
- (8) Let f be analytic on the unit disc $D(0, 1)$.
 - (a) If $f(1/n) = 1/n^2$ for all $n \geq 2$, determine f .
 - (b) Is there such an f with $f(1/n) = (-1)^n/n$ for all $n \geq 2$? Justify your answer.
- (9) Suppose $f(z) = \sum a_n z^n$ has radius of convergence $r > 0$ and $f(z) = f(-z)$ for all $|z| < r$. Show that $a_n = 0$ for all odd n .
- (10) Compute $\int_{\gamma} \bar{z} dz$ where γ is the unit circle traversed once counterclockwise, and also where γ is the boundary of the square with vertices $0, 1, 1 + i, i$. Why does this not contradict the Cauchy–Goursat theorem?
- (11) Let C_R denote the circle $|z| = R$ traversed once counterclockwise, with $R > 1$. Use the ML-estimate to show

$$\left| \int_{C_R} \frac{dz}{z^2 + 1} \right| \leq \frac{2\pi R}{R^2 - 1},$$
 and conclude that the integral tends to 0 as $R \rightarrow \infty$.
- (12) State the Cauchy–Goursat theorem carefully. In its proof, explain (i) why the nested rectangles shrink to a single point $z_0 \in R$, (ii) exactly where the hypothesis that f is $\mathbb{C}$ -differentiable at

z_0 is used, and (iii) how the estimate $|\int_{\partial R} f| \leq 4^n |\int_{\partial R_n} f|$ combines with the ML-estimate to finish the argument.